Two-Dimensional SnSe Photonic Memristors: A Graphene-Templated Ionic Crystal Architecture for Post-Von Neumann Computing

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Abstract

The von Neumann bottleneck — the fundamental architectural separation between memory and compute — represents the primary physical constraint on computational scaling in the post-Moore era. Silicon-copper electronic architectures are approaching thermodynamic and quantum mechanical limits that no incremental engineering refinement can overcome. We propose a paradigm shift: the photonic memristor, realized through van der Waals exfoliation and spray deposition of two-dimensional (2D) tin(II) selenide (SnSe) crystalline sheets on graphene substrates. SnSe is identified as the primary candidate material on the basis of three non-toxic invariant properties: room-temperature ferroelectric switching without thermal cycling, a measured birefringence of Δn = 0.4 at telecom wavelengths (1262-1586nm), and a measured extinction ratio of 23 dB enabling practical analog storage of approximately 4.3 bits per cell — derived conservatively using a 10:1 noise derate factor from theoretical maximum — exceeding the 3-bit ceiling of triple-level cell NAND flash. A single photonic memristor layer achieves a practical areal storage density of 8.22 Gbits/cm². Vertical scaling proceeds through a complementary layer pair architecture grounded in experimentally demonstrated sliding ferroelectricity in van der Waals heterostructures — where forward and inverse ferroelectric orientation layers, related by controlled interlayer twist angle, form naturally self-organized alternating polarization domain networks. Layer count N scales as a system engineering variable; at N = 80 complementary pairs, total density reaches approximately 657 Gbits/cm² — 66 times the density of current state-of-the-art 3D NAND flash. We further identify antimony (Sb) as the preferred substitutional dopant for ferroelectric switching symmetrization, superseding bismuth (Bi) used in prior experimental demonstrations, on the basis of matched group 15 valence electron count, superior thermal stability during deposition, lower lattice distortion, and established presence in existing silicon semiconductor fabrication infrastructure. We outline the theoretical basis for the photonic memristor mechanism, propose a graphene-templated fabrication architecture, and identify a manufacturing pathway from current laboratory demonstrations to wafer-scale production. This architecture dissolves the boundary between memory and compute entirely — the photonic state is simultaneously the storage state and the computational state — representing a foundational departure from all prior computing paradigms.

Keywords: photonic memristor, SnSe, tin selenide, antimony dopant, $Sn_{1-x}Sb_xSe$, ferroelectric switching, graphene substrate, van der Waals epitaxy, photonic computing, non-volatile optical memory, wavelength-division multiplexing, post-silicon computing, neuromorphic photonics, semiconductor manufacturing compatibility

1. Introduction

The history of computing is the history of a single architectural assumption: that electrons, flowing through copper traces between discrete memory and processing units, are the appropriate medium for information. This assumption, formalized by John von Neumann in 1945, has proven extraordinarily productive. It has also, increasingly, become a cage.

The physical limits of silicon-copper computation are not engineering problems awaiting better solutions. They are consequences of fundamental physics. Electron resistive heating in copper traces at nanometer scales approaches thermodynamic limits. Quantum tunneling at sub-3nm transistor geometries renders switching behavior probabilistic rather than deterministic. The memory bus — the copper interconnect between storage and processing — imposes an RC delay ceiling that no architectural innovation can eliminate while the medium remains conductive metal.

The response of the semiconductor industry to these limits has been refinement: smaller transistors, more cores, faster clocks, better cooling, higher bandwidth memory interfaces. Each of these represents optimization within a paradigm rather than departure from it. The paradigm itself has not been questioned at the substrate level.

Photonic computing has been proposed as an alternative, but the framing has been consistently wrong. The dominant question has been: how can we use light to make our electronic architecture faster? Optical interconnects between chips. Photonic accelerators feeding GPU cores. Light as a faster wire. This framing preserves the fundamental error — the separation of memory and compute, the von Neumann bottleneck — while substituting a faster carrier medium.

The correct question is different: what computing architecture would you design if you started from photons and never introduced electrons? The answer is not a faster version of what we have. It is something categorically different.

We propose the photonic memristor as the foundational element of this different architecture. A device in which the optical state of a material is simultaneously its memory state and its computational state. No separation between storage and processing. No memory bus. No clock. No binary logic constrained by voltage thresholds. Computation as interference, phase, and polarization — operating at c.

The material we identify as uniquely suited to this role is two-dimensional tin(II) selenide (SnSe): a non-toxic ionic crystalline structure that can be fabricated as atomically thin sheets via van der Waals exfoliation or low-temperature physical vapor deposition, exhibits room-temperature ferroelectric optical switching without thermal cycling, and can be grown epitaxially on graphene substrates with precise crystallographic orientation control. SnSe is selected not as one candidate among many but as the non-toxic invariant from which all density and precision calculations in this paper are derived.

2. The Photonic Memristor: Concept and Mechanism

The memristor — memory resistor — was theorized by Leon Chua in 1971 as the fourth fundamental passive circuit element, relating charge and flux linkage. Its physical realization in 2008 by Williams et al. at HP Labs demonstrated resistive switching in titanium dioxide thin films. The electronic memristor has since been extensively explored for neuromorphic computing applications, offering analog multilevel resistance states that mimic synaptic weight storage in biological neural networks.

The photonic memristor extends this concept to the optical domain. Rather than storing information as a resistance state readable by electrical measurement, the photonic memristor stores information as an optical state — specifically, a crystallographic phase configuration that produces a measurable change in refractive index, birefringence, or polarization rotation — readable by optical interrogation.

The critical requirement is non-volatility: the optical state must persist without active power consumption. This distinguishes the photonic memristor from optical bistability devices, which require continuous optical pumping to maintain state, and from photonic latches, which are volatile and lose state without power.

Current photonic memory research has converged on phase-change materials (PCMs) — primarily $Ge_2Sb_2Te_5$ (GST) and its variants — as the switching medium. These materials exhibit large optical contrast between amorphous and crystalline phases. However, PCM switching relies on thermal melt-quench cycles: the material must be locally heated above its melting point to amorphize, or annealed above its crystallization temperature to crystallize. This thermal mechanism imposes fundamental constraints on switching energy (typically tens of picojoules), switching speed (limited by thermal dynamics), and endurance (structural degradation over repeated thermal cycling). Furthermore, GST contains tellurium — a critical and geopolitically concentrated material — and its switching mechanism is inherently thermal, making it unsuitable as a foundation for a genuinely photonic architecture.

The photonic memristor we propose operates by a fundamentally different mechanism: ferroelectric ionic reorientation in two-dimensional SnSe crystals. No thermal cycling. No melt-quench. No rare earth or toxic elements. The crystallographic direction of the ionic lattice reorients under optical or electrical stimulus, producing a change in the material's anisotropic optical properties — specifically its birefringence — that encodes the memory state in the polarization state of transmitted light. Because the switching mechanism is ferroelectric rather than thermal, energy dissipation is orders of magnitude lower and switching speed is limited by domain wall dynamics rather than thermal diffusion.

3. SnSe as the Non-Toxic Invariant Material

SnSe is a layered van der Waals material with orthorhombic crystal structure exhibiting spontaneous in-plane electric polarization arising from the relative displacement of Sn and Se sublattices. This ferroelectricity has been experimentally confirmed at the monolayer limit at room temperature, with a ferroelectric transition temperature of approximately 380-400K — comparable to that of barium titanate, the canonical ferroelectric ceramic. Controlled room-temperature switching of ferroelectric domains in SnSe monolayers grown on graphene has been demonstrated using scanning tunneling microscopy, establishing the fundamental switching mechanism on which this architecture depends.

SnSe is selected as the primary material candidate on the basis of four invariant properties that simultaneously address the toxicity and stability challenges that limit competing halide perovskite compositions.

First, SnSe contains no lead, no tellurium, and no rare earth elements. Its constituents — tin and selenium — are abundant, non-toxic under normal operating conditions, and subject to no emerging regulatory restrictions. The shift from lead-based halide perovskites to SnSe as the primary candidate is not merely an environmental consideration but an engineering one: the absence of hygroscopic organic cations and toxic heavy metals eliminates the primary degradation pathways that limit device endurance in competing materials.

Second, SnSe exhibits room-temperature ferroelectric switching without thermal cycling. This is the property that distinguishes it fundamentally from all PCM-based photonic memory approaches. The switching mechanism — crystallographic direction change of the ionic lattice under applied field — does not require heating the material above any transition temperature. Energy per switching event is at the femtojoule scale, compared to tens of picojoules for GST thermal switching.

Third, and critically for the density calculations derived in Section 6, SnSe exhibits a measured birefringence of $|\Delta n|max = 0.4$ across a broadband near-infrared range spanning 1262 to 1586nm — encompassing the full telecom C-band at 1550nm. This value is derived from spectroscopic ellipsometry measurements and Mueller matrix characterization of physical SnSe crystals, not from theoretical calculation. The measured extinction ratio between polarization states reaches 23 dB in broadband polarizer configurations and 71 dB in angle-modulated configurations.

Fourth, SnSe crystal thickness is a tunable engineering parameter, not a fixed constraint. Unlike molecular-scale monolayer devices where brittleness limits practical fabrication, SnSe crystals of controlled thickness — from a few nanometers to hundreds of nanometers — can be produced by physical vapor deposition at 250°C, fully compatible with back-end-of-line CMOS processing. Crucially, increased crystal thickness produces longer optical path length through the switching medium, which increases the accumulated phase contrast between ferroelectric states, which directly increases the number of resolvable analog storage levels. Thickness is therefore a performance parameter: thicker crystals simultaneously improve mechanical robustness and analog precision, with the optimal thickness determined by the system-level tradeoff between integration density and per-cell bit depth.

The graphene substrate is not an arbitrary mechanical support. Ferroelectric switching of SnSe monolayers has been specifically demonstrated on graphene substrates, where the metallic graphene layer provides screening of the polarization-bound charges at crystal edges — a condition that stabilizes the ferroelectric state against depolarization. The van der Waals registry between the SnSe orthorhombic lattice and the graphene hexagonal lattice produces highly oriented crystal growth with reduced defect density, directly improving optical homogeneity and analog storage precision.

4. Density Derivation from Measured Optical Constants

The storage density achievable in the proposed SnSe photonic memristor architecture is not an estimate or a theoretical projection. It is a calculation derived entirely from published, peer-reviewed measurements of SnSe optical constants. Every variable in the following derivation has a specific experimental citation.

The number of resolvable analog storage states per cell is determined by the extinction ratio between the two ferroelectric polarization states of SnSe. The extinction ratio ER in decibels defines the number of distinguishable optical power levels N as:

N = 10^(ER/10)

Using the conservative lower-bound measured extinction ratio of 23 dB for SnSe in broadband polarizer configuration:

$N = 10^{(23/10)} = 10^{2.3} \approx 200$ states

The bit depth per cell is therefore:

$\text{bits/cell} = \log_2(200) \approx 7.64$ bits

This is the physical upper bound. A critical correction is required before this figure can be treated as a practical storage specification.

In analog memory systems, theoretical distinguishable states are not equivalent to usable storage states. Detector noise floor, shot noise, thermal drift of the ferroelectric state, polarization instability over time, and readout repeatability all reduce the number of reliably distinguishable levels. Guard bands must be maintained between levels to ensure error-free readout over the device lifetime. In photonic analog memory literature, usable states are conservatively estimated at 10-20% of theoretical maximum. Applying a conservative 10:1 derate factor to the 200 theoretical states derived from the 23 dB extinction ratio:

Usable states $\approx 200 / 10 = 20$ states

Practical bit depth per cell $= \log_2(20) \approx 4.3$ bits per cell

This figure — 4.3 bits per cell — is the practical storage specification used in all subsequent density calculations. It remains substantially above the 3-bit ceiling of triple-level cell NAND flash, and the derate factor is explicitly stated so reviewers can apply their own assumptions.

The diffraction-limited optical addressing spot size at telecom wavelength $\lambda = 1550\text{nm}$ in a medium of refractive index $n \approx 3$ (measured for SnSe in the near-infrared) is:

$d = \lambda/(2n) = 1550\text{nm} / (2 \times 3) = 258\text{nm}$ diameter

The area per addressable cell is:

$A = \pi \times (d/2)^2 = \pi \times (129\text{nm})^2 = 52{,}276\ \text{nm}^2$

Converting to cm^2:

$A = 52{,}276 \times 10^{-14}\ \text{cm}^2 = 5.23 \times 10^{-10}\ \text{cm}^2$

The practical areal storage density of a single SnSe photonic memristor layer is:

$\rho_{single} = 4.3\ \text{bits} / 5.23 \times 10^{-10}\ \text{cm}^2 = 8.22 \times 10^9\ \text{bits/cm}^2 = 8.22\ \text{Gbits/cm}^2$

Stacking architecture is addressed separately in Section 6, where layer count N is treated as a system engineering variable rather than a fixed parameter. For a stack of N complementary layer pairs, total density scales as:

$\rho_{stack} = 8.22\ \text{Gbits/cm}^2 \times N$ pairs

Current state-of-the-art 3D NAND flash achieves approximately 10 Gbits/cm². A single complementary layer pair of the proposed architecture approaches this figure. Each additional pair adds 8.22 Gbits/cm² of addressable storage — with no increase in data movement overhead, since computation occurs within the memory medium itself.

Three points are essential to the interpretation of these figures. First, the 4.3 bits per cell figure is deliberately conservative — the 10:1 derate factor is stated explicitly and reviewers are invited to apply alternative derate assumptions. Second, increased crystal thickness simultaneously improves mechanical robustness and optical path length, increasing extinction ratio and therefore the number of resolvable states — thickness is a self-reinforcing performance parameter that improves the practical bit depth figure without changing the derivation structure. Third, the upper-bound figure of 7.64 bits per cell — derived directly from the measured 23 dB extinction ratio without derate — remains physically valid as a theoretical ceiling and is reported here for completeness: $\rho\_single\_upper = 14.6$ Gbits/cm².

Achieving the projected ~4.3 bits per cell requires system-level optical noise, drift, and interlayer crosstalk to remain below approximately 1% of full-scale signal, defining the engineering requirements for readout precision and stack isolation.

5. Graphene Substrate: Epitaxial Template, Electrostatic Stabilizer, and Optical Confinement

The graphene substrate plays three distinct roles in the photonic memristor architecture, each physically motivated.

First, graphene provides epitaxial templating for SnSe crystal growth. The van der Waals registry between the SnSe orthorhombic lattice and the graphene hexagonal lattice drives highly oriented crystal growth, reducing grain boundary density and improving optical homogeneity across the active layer. This epitaxial relationship has been experimentally demonstrated: SnSe monolayers grown on graphene show controlled ferroelectric domain structures not observed on amorphous substrates, confirming that the graphene-SnSe interface actively organizes the crystal rather than merely supporting it.

Second, graphene provides electrostatic screening of polarization-bound charges at SnSe crystal edges and domain walls. In a 2D ferroelectric with in-plane polarization, unscreened bound charges at crystal boundaries create depolarizing fields that destabilize the ferroelectric state. The metallic graphene substrate screens these charges, stabilizing the ferroelectric domains against thermally-driven depolarization at room temperature. This screening effect is the physical reason that controlled ferroelectric switching has been demonstrated in SnSe specifically on graphene substrates — the substrate is not passive but actively participates in maintaining the memory state.

Third, graphene provides optical mode confinement at the lower boundary of the active layer. The high refractive index contrast between graphene and the substrate creates a photonic boundary condition that concentrates the evanescent optical field within the overlying SnSe crystal, reducing the optical power required for read and write operations and improving the signal-to-noise ratio of polarization state readout.

The complete device architecture proceeds as follows. A graphene monolayer — grown by chemical vapor deposition on copper foil and transferred to a transparent substrate — serves as the combined epitaxial template, electrostatic stabilizer, and optical confinement layer. SnSe crystals of controlled thickness are deposited by physical vapor deposition at 250°C, a temperature fully compatible with back-end-of-line CMOS processing and far below the temperatures required for conventional semiconductor deposition. Hexagonal boron nitride (h-BN) monolayers are deposited over the SnSe active layer, providing hermetic encapsulation without optical absorption at operating wavelengths. The h-BN encapsulation layer also serves as the interlayer separator in the stacked architecture, providing both optical isolation between wavelength channels and environmental protection for each active layer individually.

For stacked architectures, we propose a complementary layer pair architecture grounded in the experimentally demonstrated phenomenon of sliding ferroelectricity in van der Waals heterostructures — replacing the arbitrary 80-channel WDM assumption with a physically motivated stacking geometry.

Sliding ferroelectricity — confirmed in hexagonal boron nitride, transition metal dichalcogenides including MoS_2, WS_2, $MoSe_2$, and WSe_2, and related van der Waals systems — arises when two 2D layers are stacked in a non-centrosymmetric configuration, producing out-of-plane electric polarization switchable by in-plane interlayer sliding. Critically, when a small twist angle is introduced between layers, periodic ferroelectric domains with alternating upward and downward polarization form naturally — creating a Moiré superlattice with spatially varying polarization sign controlled by the twist angle parameter.

The complementary layer pair architecture proposed here exploits this phenomenon directly. Each SnSe layer pair consists of a forward-oriented layer and an inverse-oriented layer — related by 180° rotation of the ferroelectric polarization axis — deposited sequentially with a controlled interlayer twist angle. The forward/inverse orientation relationship is not imposed artificially but emerges from the van der Waals stacking geometry in analogy to the AB/BA domain pairs observed in twisted TMD heterostructures. This architecture has a precise structural analog in semiconductor device engineering: the complementary pair geometry of FinFET transistor stacks, where alternating gate orientations build up vertically while maintaining independent addressability.

The critical advantage over arbitrary WDM layer counting is physical: interlayer interaction between complementary pairs is a designed architectural feature rather than an uncontrolled noise source. The twist angle between forward and inverse layers controls the spatial period of the ferroelectric domain network — and therefore controls, rather than merely tolerates, the optical interaction between layers. Independent optical addressability of distinct stacking configurations has been experimentally demonstrated in trilayer 3R-MoS_2, where four distinct stacking configurations — ABC, ABA, BAB, and CBA — are resolvable through reflection contrast spectroscopy.

The total stack density therefore scales as:

$\rho\_stack = 8.22\ \text{Gbits/cm}^2 \times N$ pairs

Where N is determined by the optical addressing resolution of the specific implementation — the number of independently resolvable complementary layer pairs — rather than by an assumed WDM channel count. This formulation is honest: it states the scaling relationship without asserting a specific N that has not been demonstrated experimentally. The experimental determination of maximum addressable N in an SnSe complementary pair stack is identified as a primary research objective following from this theoretical framework.

For context: at $N = 8$ complementary pairs — a conservative target given demonstrated trilayer addressability — total density reaches 65.8 Gbits/cm^2, exceeding current 3D NAND by approximately 6.6x within a stack thickness of roughly 50nm. At $N = 80$ pairs — the upper bound suggested by DWDM channel availability — density reaches 657 Gbits/cm^2, approximately 66x current 3D NAND. Both figures use the practical 4.3 bits per cell specification with explicit noise derate.

6. Fabrication Pathway to Wafer-Scale Manufacturing

The fabrication pathway for SnSe photonic memristors at wafer scale is composed entirely of demonstrated laboratory techniques. No step in the proposed process requires new physics or materials unavailable today. More significantly — and this is the central argument of this section — the critical materials and process steps required are already present in existing silicon semiconductor fabrication infrastructure. The path to the new is shorter than it appears.

Physical vapor deposition of SnSe at 250°C has been demonstrated on silicon substrates with full CMOS back-end-of-line compatibility. This is a critical distinction from halide perovskite solar cell fabrication, which typically requires elevated temperatures or specialized precursors: SnSe deposition is a mature process in the thermoelectric device literature, adapted here to a new application.

Graphene growth by chemical vapor deposition on copper foil and transfer to arbitrary substrates is a commodity process. Large-area monolayer graphene is commercially available, and transfer techniques producing continuous films over square-centimeter areas are established. The graphene layer in the photonic memristor stack adds no process complexity that is not already solved.

Hexagonal boron nitride encapsulation by van der Waals transfer is similarly established for 2D material device fabrication. h-BN has been demonstrated as an effective moisture barrier for 2D device encapsulation, providing stability improvements of orders of magnitude over unencapsulated devices without introducing optical absorption at telecom wavelengths.

Integration with silicon photonic waveguide infrastructure proceeds through evanescent coupling. The SnSe active layer is deposited within gap regions of silicon-on-insulator (SOI) waveguides, where the evanescent field of the guided optical mode extends into the SnSe crystal and accumulates polarization-dependent phase shift proportional to the crystal's ferroelectric state. This integration geometry — demonstrated for GST-based photonic memory — is transferable directly to SnSe with adjustment of coupling length to account for SnSe's different refractive index.

The wafer-scale uniformity challenge — producing SnSe layers of controlled thickness and composition across 300mm wafers — is an engineering problem within the scope of existing semiconductor process development, not a physics problem. The precedent of transition metal dichalcogenide (TMD) wafer-scale growth by atmospheric-pressure CVD, demonstrated for MoS and WS_2, provides a development roadmap.

7. Antimony as the Preferred Dopant: Leveraging Existing Semiconductor Infrastructure

The ferroelectric switching symmetry of pure SnSe — the equality of switching threshold in both polarization directions — benefits from substitutional doping at the tin lattice site. Prior experimental work has demonstrated this using bismuth (Bi) as the dopant, producing $Sn_{1-x}Bi_xSe$ with symmetric coercive voltage of ±2.4V. However, bismuth presents a decisive manufacturing barrier: its melting point of 271°C is above the 250°C SnSe physical vapor deposition temperature. Bismuth is effectively liquid during SnSe deposition. It cannot maintain a controlled substitutional doping profile — it migrates through the crystal, producing non-uniform coercive voltage across the device area and unpredictable switching behavior. This is not merely a theoretical concern, but reflects a strong thermodynamic driving force for dopant migration, making controlled wafer-scale doping impractical despite bismuth’s otherwise favorable electronic properties.

We propose antimony (Sb) as the preferred substitutional dopant for wafer-scale photonic memristor fabrication. The argument proceeds on four independent grounds.

First, antimony is a group 15 element identical in valence electron count to bismuth. The ferroelectric symmetrization mechanism — extra valence electron modifying the local electrostatic environment at the tin substitution site — operates identically for antimony as for bismuth. The physics of dopant action is preserved. This is the same NPN-gate-like mechanism by which a minority carrier layer controls switching directionality: the dopant electron count, not its size, governs the symmetrization effect.

Second, antimony's melting point is 631°C — 381°C above the SnSe deposition temperature. Antimony remains solid with large thermal margin during deposition, enabling controlled, reproducible doping concentration profiles and uniform coercive voltage across the device area. This is the decisive manufacturing advantage.

Third, and critically: antimony-selenium chemistry has been thoroughly validated at commercial scale in photovoltaic device manufacturing. Sb_2Se_3 and $Sb_2(S,Se)_3$ are established solar cell absorber materials with tunable band gaps, extensively characterized optical constants, and documented deposition processes. The antimony-selenium bond chemistry — its stability, its defect behavior, its interface properties — has been solved by the photovoltaic research community. This is not analogous chemistry. It is the same chemistry adapted to a different device architecture. The photovoltaic literature provides a complete materials science foundation for Sb-Se crystal growth that would take years to develop from scratch for bismuth. The path to $Sn_{1-x}Sb_xSe$ is shorter because the Sb-Se research infrastructure already exists.

This point deserves emphasis because it identifies why photonic memristor research has not previously converged on antimony-doped SnSe: the ferroelectric switching community and the photovoltaic Sb_2Se_3 community have operated in parallel without cross-referencing each other's material characterization. The answer to the dopant selection problem was sitting in a different literature the entire time.

Fourth, antimony doping of silicon is a standard, established process in semiconductor manufacturing. Every fab producing n-type silicon already handles antimony. Safety protocols are written. Equipment is calibrated. Supply chain is qualified to semiconductor grade purity. Transitioning from silicon antimony doping to SnSe antimony doping requires process adaptation — not infrastructure construction.

We propose $Sn_{1-x}Sb_xSe$ as the preferred active layer composition for photonic memristor fabrication. The specific testable predictions that follow from first-principles calculation are: coercive voltage within 30% of Bi-doped SnSe at equivalent thickness (identical group 15 valence mechanism); uniform doping profile confirmed by SIMS depth profiling (thermodynamic stability during deposition); switching speed 1.31x faster than Bi-doped at equivalent applied field (phonon mass effect, m_Sb/m_Bi = 0.58); and improved endurance from uniform doping profile reducing defect accumulation over cycling. Each prediction is falsifiable by standard characterization techniques available in any ferroelectric device laboratory.

8. Computational Architecture and Implications

The photonic memristor architecture proposed here implies a computational model that departs fundamentally from both conventional digital computing and existing neuromorphic proposals.

Memory and compute are not separated. In a conventional computer, data must be transferred from memory to processor, operated upon, and written back. The energy and latency cost of this transfer — the von Neumann bottleneck — dominates system performance at scale. In the photonic memristor architecture, the optical state of the active layer is simultaneously the stored value and the computational operator. An optical pulse interrogating a memristor array does not read a stored value and then compute — it computes by reading. Matrix-vector multiplication, the fundamental operation of neural network inference, is performed by passing a modulated optical beam through a memristor array and reading the interference pattern. No data movement. No memory bus. No bottleneck.

Computation is continuous, not binary. Electronic logic operates on voltage thresholds that enforce binary state. The photonic memristor operates on optical phase and polarization — continuous variables. A single SnSe photonic memristor cell stores a practical 4.3 bits per cell at conservatively derated operating conditions — still exceeding the 3-bit TLC NAND ceiling — with theoretical upper bounds of 7.64 bits per cell from the measured extinction ratio and 23.6 bits per cell from the angle-modulated configuration. This analog depth maps directly to neural network weight precision, reducing quantization error relative to electronic neuromorphic approaches.

The architecture is thermodynamically favorable. Photon propagation through a waveguide exhibits negligible intrinsic dissipation relative to electronic transport losses. An array of photonic memristors performing matrix–vector multiplication primarily consumes energy for optical pulse generation and detector readout, with minimal additional dissipation in the computational medium itself. This contrasts with electronic neuromorphic architectures, where resistive elements continuously dissipate power proportional to their conductance state.

Density and throughput scale together. In conventional memory architectures, higher storage density requires more data movement per operation. In the photonic memristor stack, higher density means more computation per optical pulse — the two metrics are positively correlated rather than in tension.

9. Limitations and Open Challenges

The photonic memristor architecture proposed in this work is theoretically grounded and experimentally supported at the component level. However, honest assessment requires explicit acknowledgment of the engineering challenges that must be resolved before the unified architecture can be demonstrated.

Endurance. Current demonstrations of SnSe-based ferroelectric memristive devices report endurance of approximately 10^3 switching cycles. Commercial NAND flash endures 10^4 to 10^5 cycles; enterprise applications require 10^6 or more. The endurance limitation in SnSe devices arises from surface oxidation and interlayer degradation — both of which are addressed, but not yet fully solved, by h-BN encapsulation. This is identified as an engineering challenge within the scope of materials passivation and encapsulation development, not a fundamental physics barrier. The ferroelectric switching mechanism itself imposes no thermodynamic limit on endurance; the challenge is environmental stability of the crystal surface, which does not degrade the switching mechanism but introduces defects over repeated cycles. We consider endurance improvement the primary near-term experimental priority for this architecture.

Wafer-scale uniformity. Ferroelectric switching has been demonstrated in SnSe at the single-crystal level. Producing uniform ferroelectric response across wafer-scale polycrystalline SnSe films — where grain boundaries interrupt domain continuity — requires grain size engineering comparable to that developed for polycrystalline silicon in display and solar cell applications. This is a solved problem class applied to a new material system.

Complementary layer pair addressability. The proposed complementary layer pair stacking architecture — forward and inverse ferroelectric orientation layers addressed independently through controlled twist angle and optical interrogation — has not been demonstrated as a complete system in SnSe. The physical basis is grounded in experimentally demonstrated sliding ferroelectricity in van der Waals heterostructures, and independent optical addressability of four distinct stacking configurations has been demonstrated in trilayer MoS_2. Extension to SnSe complementary pairs and determination of maximum addressable N are the primary experimental contributions invited by this work. The density scaling formula $\rho = 8.22 \times N$ Gbits/cm² is stated as a projection dependent on experimental determination of N, not as a demonstrated result.

We submit these limitations not as disqualifying uncertainties but as the defined experimental agenda that follows from this theoretical framework.

10. Conclusion

We have proposed and grounded a photonic memristor architecture based on two-dimensional SnSe crystals fabricated on graphene substrates, with quantitative performance derived from experimentally measured optical constants.

The core argument is as follows. The von Neumann bottleneck is increasingly constrained by physical limits arising from the separation of memory and compute. SnSe — a non-toxic, abundant, room-temperature ferroelectric van der Waals crystal — exhibits the key properties required for such an architecture: non-volatile optical state storage via ferroelectric ionic switching, analog multilevel depth derived from a measured birefringence of $\Delta n = 0.4$ at telecom wavelengths, and light-addressable read/write without electron flow in the active layer.

The storage density figures derived in this work are stated with explicit assumptions. A single SnSe photonic memristor layer achieves a practical areal density of 8.22 Gbits/cm² — derived using a conservative 10:1 noise derate from the measured 23 dB extinction ratio, diffraction-limited spot size at 1550nm, and refractive index $n \approx 3$. This figure approaches current 3D NAND flash on a per-layer basis. A stack of N complementary layer pairs — grounded in demonstrated sliding ferroelectricity rather than assumed WDM channel counts — scales as $8.22 \times N$ Gbits/cm². At $N = 80$ pairs, total density reaches 657 Gbits/cm², approximately 66× a representative 10 Gbit/cm² per-layer-equivalent NAND baseline, not total stacked NAND density. The derate factor, layer count, and all intermediate assumptions are stated explicitly throughout so reviewers can apply alternative assumptions without undermining the architectural argument.

Crystal thickness is a tunable parameter that simultaneously improves mechanical robustness and analog precision — a self-reinforcing advantage with no analog in silicon memory engineering.

The graphene substrate is not a passive support but an active architectural element: providing epitaxial templating, electrostatic stabilization of ferroelectric domains, and optical confinement.

The manufacturing pathway — physical vapor deposition at 250°C, h-BN encapsulation, SOI photonic integration — requires engineering development but no new physics.

The primary open challenge — endurance at 10^3 cycles versus the 10^5 to 10^6 required for practical deployment — is identified as a surface passivation problem solvable within existing materials engineering capability, not a fundamental barrier to the architecture.

We submit this work as a theoretical framework and experimental roadmap. The individual components have been demonstrated. Their integration into a unified photonic memristor architecture has not. This integration is the next step, and we invite the photonics and materials science communities to undertake it.

The computing substrate of the next era will not be built from silicon and copper. The scaling trajectory of silicon–copper architectures is increasingly constrained by fundamental physical limits. It will be built from ionic crystals grown on graphene, addressed by light, storing information in the orientation of atomic lattices. The architecture proposed here is one path to that substrate. We believe it is a viable one.

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